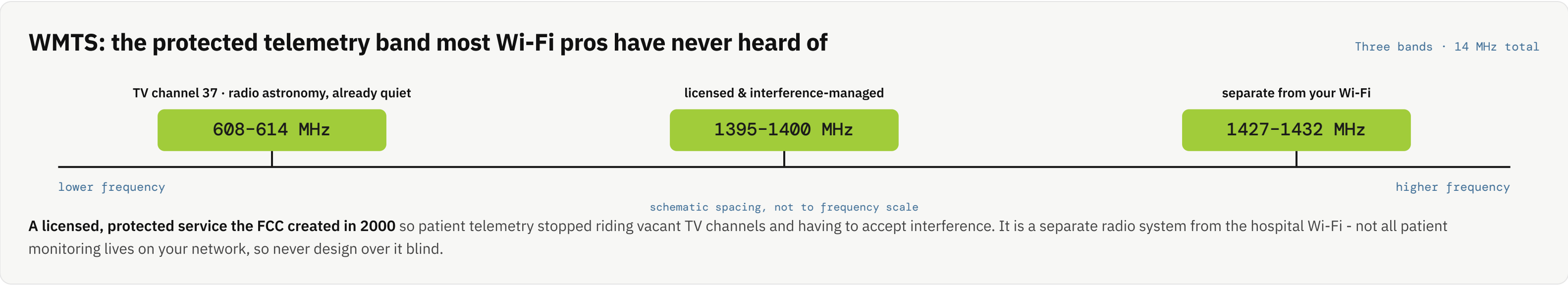


Why you cannot design a hospital like an office

A hospital is the one building where a Wi-Fi mistake can contribute to a patient-safety event - life-critical traffic, a protected telemetry band, EMC-regulated devices sharing the air, and a stack of authorities no single WLAN cert teaches.



WHY THE OFFICE PLAYBOOK FAILS

(o)

Zero-tolerance roaming

A dropped association or a slow re-auth is a **missed alarm**, not a buffering annoyance. Clinicians, mobile carts, and telemetry must roam **seamlessly facility-wide with no dead zones** - stairwells, elevators, and back-of-house included. Overlapping cells, fast-roaming (802.11r/k/v), voice-grade RF.

📍

RTLS and telemetry multiply AP count

Designs are graded **Data → Voice → Voice/RTLS**, each needing more APs. Voice/RTLS grade (location to ~5 m) demands far higher density than data grade - quote a hospital at data-grade density and it fails. RTLS of assets, staff, and patients often rides the same infrastructure.

📶

Medical-device EMC

Devices are tested to tolerate a defined RF environment under **IEC 60601-1-2** - "tested to a limit" is not "immune to anything." A dense, high-EIRP footprint can push local field strength past what a nearby monitor or pump was qualified for. Coexistence awareness; route EMC to clinical engineering.

THE FOUR-AUTHORITY STACK (RECOGNIZE AND DEFER)

HIPAA · HHS

Data: PHI confidentiality, integrity, and transmission security.
Design for it (encryption, access control, audit logging, segmentation). *defer to the privacy / security officer*

FDA

Wireless inside the medical device: QoS, coexistence, security, EMC.
Provide a sane RF environment. *defer device qualification to the maker + biomed*

FCC

Spectrum: WMTS licensing / coordination; unlicensed 2.4/5/6 GHz rules.
Respect WMTS as protected; design Wi-Fi within Part 15. *WMTS coordinator*

TJC

Accreditation: Environment of Care, alarm-management, ICRA/ILSM in construction.
Coordinate; construction access via the codes & safety reference. *the AHJ + facility*

 THE ONE HANDOFF

Coordinate with clinical / biomedical engineering first. They own the medical devices, their EMC posture, and the telemetry systems - the single most important handoff before you touch a clinical RF environment.